Ting-Yu Pan

Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign

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Education

University of Illinois Urbana-Champaign **Aug. 2026 – May 2031 (exp)**

Ph.D. in Computer Science, Advisor: Prof. Minje Kim *Urbana, IL*

Selected Courses: (In Progress) Machine Learning for Signals, Advanced Topics in NLP

University of Michigan – Ann Arbor **Aug. 2024 – May 2026**

MS in Electrical and Computer Engineering (Specialization: Machine Learning & Signal Processing) *Ann Arbor, MI*

GPA: 3.8 | Selected Courses: Computer Graphics and Generative Models, Advanced Topics in Computer Vision, Matrix Methods for Signal Processing and Machine Learning, Probability and Random Processes

National Tsing Hua University **Sept. 2019 – Jan. 2024**

BS in Power Mechanical Engineering (Minor: Computer Science) *Hsinchu, Taiwan*

GPA: 3.9 | Selected Coursework: Music Information Retrieval, Music Artificial Intelligence, Deep Learning, Design and Analysis of Algorithms

Professional Experience

University of Illinois Urbana-Champaign **Aug. 2026 – Present**

Graduate Research Assistant; Advisor: Prof. Minje Kim *Urbana, IL*

- Proposed **singer-enrollment-conditioned source extraction** using pretrained speaker embeddings, achieving **+2.2 dB SI-SDRi** on unseen vocal parts while label-based conditioning yielded negative SI-SDRi.
- Investigated the role of **training-data diversity** in transferable source separation, using voice-conversion augmentation to expand singer diversity from 8 to 43 identities and evaluating multi-corpus training and cross-corpus model selection.
- Exploring **informed and generative approaches to music source separation**, including symbolic music cues and generative source priors.
- First-author paper submitted to **ICASSP 2027** [1].

University of Michigan **May 2025 – Aug. 2026**

Graduate Research Assistant; Advisor: Prof. Hao-Wen Dong *Ann Arbor, MI*

- Developed **ACappellaSet**, a 2.5-hour multilingual dataset of 53 professionally recorded a cappella songs with isolated vocal stems and variable ensemble configurations, supporting research in source separation and music information retrieval.
- Fine-tuned **Demucs-based models** for vocal-part and vocal-percussion separation, achieving a **2.4 dB SDRi** over pretrained baselines and evaluating the impact of separation quality on downstream F0 and rhythm estimation.
- Conducted a **formative user study** of novice a cappella practice and co-designed **AcaMate**, an interactive rehearsal system integrating part-wise visualization and performance feedback to address challenges in self-diagnosis and ensemble awareness.
- Published related work at **NeurIPS 2025 AI for Music Workshop** [2], **ISMIR 2025 LBD** [3], and **UIST 2025 Adjunct** [4].

Taiwan Semiconductor Manufacturing Company (TSMC) **Jul. 2023 – Aug. 2023**

Engineer Intern, Department of Operation and System Methodology *Hsinchu, Taiwan*

- Deployed a **CNN-based defect detection pipeline** in PyTorch for automated visual inspection, achieving **88% accuracy** across semiconductor manufacturing lines.
- Automated large-scale data reporting workflows in Python, reducing manual processing by **1,100+ hours/year** across 6+ production teams; received the **Contribution Award**.

Research Center for Information Technology Innovation, Academia Sinica **Jul. 2022 – Dec. 2022**

Research Intern; Advisor: Prof. Yi-Hsuan Yang *Taipei, Taiwan*

- Developed **GAN-based music style transfer** models by adapting image-domain generative techniques to spectrogram representations for audio generation.

Publications & Presentations

[1] **Ting-Yu Pan**, Kexin Phyllis Ju, Christian Li, Minje Kim, Hao-Wen Dong. "ACappellaSet: A Dataset for Transferable A Cappella Source Separation." *Submitted to IEEE ICASSP 2027.*

[2] **Ting-Yu Pan**, Kexin Phyllis Ju, Hao-Wen Dong. "ACappellaSet: A Multilingual A Cappella Dataset for Source Separation and AI-Assisted Rehearsal Tools." *NeurIPS 2025 Workshop on AI for Music.*

[3] **Ting-Yu Pan**, Kexin Phyllis Ju, Hao-Wen Dong. "Curating an A Cappella Dataset for Source Separation." *Late-Breaking Demo, ISMIR 2025.*

[4] Kexin Phyllis Ju, **Ting-Yu Pan**, Hao-Wen Dong. "AcaMate: Supporting Novice A Cappella Singers in Individual Practice." *UIST 2025 Adjunct.*

Selected Projects

DeepMUSHRA: Reference-Aware Audio Quality Assessment **Aug. 2026 – Present**

Research Project *Urbana, IL*

- Developing a **reference-aware large audio-language model (LALM)** for perceptual audio quality assessment, generating interpretable descriptions of subtle differences between reference and processed audio.
- Designed a **synthetic audio degradation pipeline** with controlled codec-style artifacts and metadata-derived supervision, with evaluation against expert MUSHRA ratings on ODAQ.

Symbolic Music Generation **Feb. 2022 – Feb. 2023**

Undergraduate Research; Advisor: Prof. Chuan-Kang Ting *Hsinchu, Taiwan*

- Developed a **conditional generative model for symbolic a cappella arrangement** using CoCoNet, generating three vocal accompaniment parts from a given lead melody through masked modeling and iterative Gibbs sampling.
- Conducted a **blinded listening study with 160 participants** comparing machine- and human-arranged music across perceptual quality, harmony, and arrangement richness; 30–40% rated generated arrangements as comparable to or better than human arrangements.

Awards & Honors

Empowering Research with AI Award – First Prize ($3,000), University of Michigan AI Institutes 2026

Contribution Award, Taiwan Semiconductor Manufacturing Company (TSMC) 2023

Awarded for outstanding performance and engineering contributions among departmental interns.

Academic Achievement Award, National Tsing Hua University 2023

Awarded for outstanding academic performance, ranking in the top 5% of the department.

Music Experience

Professional A Cappella Artist **2015 – Present**

Performer, Arranger, Composer & Vocal Coach *[portfolio]* *Taiwan & International*

- Core member of **Seagull-K**, a professional contemporary a cappella group; performed in **500+ concerts and televised showcases** across multilingual repertoire.
- Co-arranged and contributed to original and culturally grounded compositions recognized by the **Hakka Pop Music Awards** and **Dongfeng Music Festival**.
- Coached student ensembles and led music-education initiatives, including youth vocal training programs, workshops, and regional performance programs.

Academic Service & Teaching

NeurIPS 2025 Workshop on AI for Music **2025**

Program Committee Reviewer

University of Michigan **Sep. 2025 – Dec. 2025**

Grader, ECE 551: Matrix Methods for Signal Processing, Data Analysis and Machine Learning *Ann Arbor, MI*

References

Prof. Minje Kim

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minje@illinois.edu